CLAIMS

What is claimed is:

1. A method for routing data over a massively parallel optical interconnect having a two-dimensional array of emission channels, the method comprising the steps of:

    - receiving a plurality of data packets, wherein each data packet is associated with a network destination from a set of possible destinations;
    - assigning a unique emission characteristic to each network destination in the set;
    - for each data packet, determining its associated network destination and identifying the corresponding unique emission characteristic;
    - generating a data frame by assigning the data from said data packets to a plurality of channels in the two-dimensional array and setting the emission characteristic of each channel to the characteristic corresponding to the destination of the data it contains; and
    - transmitting said data frame via the optical interconnect.

2. The method of claim 1, wherein the emission channels are pixels and the emission characteristic is the color of the pixels.

3. The method of claim 2, wherein the set of possible destinations is mapped to a set of colors, and wherein the step of generating the data frame comprises coloring the pixels according to the destination of the data they carry.

4. The method of claim 1, further comprising the steps of:

    - receiving, at a receiver, the transmitted data frame;
    - for each channel in the received frame, identifying its emission characteristic; and

- routing the data from each channel to its network destination based on the identified emission characteristic.

5. The method of claim 1, wherein the steps are performed by a processor integrated into a connector housing of the optical interconnect.

6. A system for network routing, comprising:

   - a massively parallel optical interconnect including a two-dimensional array of individually addressable emission channels;
   - a processor configured to:
     - receive data streams associated with a plurality of network destinations;
     - maintain a routing table that maps each network destination to a unique emission characteristic;
     - generate a sequence of data frames by populating the channels of the array with data from the data streams and setting the emission characteristic of each channel according to the mapping in the routing table; and
   - an encoder circuit configured to drive the emission array to transmit the sequence of data frames.

7. The system of claim 6, wherein the emission channels are pixels on a micro-display and the unique emission characteristic is color.

8. The system of claim 6, further comprising a decoder circuit at a receiving end of the interconnect, the decoder configured to read the emission characteristic of each channel and to route the data contained therein to the corresponding network destination.

9. The system of claim 6, wherein the processor is a GPU, FPGA, or ASIC located within a connector head of the interconnect.

10. The system of claim 6, wherein the routing method constitutes a form of software-defined wavelength-division multiplexing.